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Methods and apparatus for performing measurements on an ultrasound image

Abstract

Aspects of the technology described herein include a processing device configured to display, on a touch-sensitive display screen of a processing device in operative communication with an ultrasound device, an ultrasound image, a movable measurement tool, and an icon that maintains a fixed distance from a portion of the measurement tool. The icon may be configured to modify the measurement tool, and the icon may not overlap the measurement tool.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation under 35 USC § 120 of U.S. patent application Ser. No. 18/141,450 entitled “METHODS AND APPARATUS FOR PERFORMING MEASUREMENTS ON AN ULTRASOUND IMAGE”, filed Apr. 30, 2023, which is a continuation under 35 USC § 120 of U.S. patent application Ser. No. 16/662,288, entitled “METHODS AND APPARATUS FOR PERFORMING MEASUREMENTS ON AN ULTRASOUND IMAGE”, filed Oct. 24, 2019, now U.S. patent Ser. No. 11/638,572, which claims priority and the benefit under 35 U.S.C. 119(e) of U.S. Provisional Patent Application Ser. No. 62/750,348, filed Oct. 25, 2018, entitled “METHODS AND APPARATUS FOR PERFORMING MEASUREMENTS ON AN ULTRASOUND IMAGE”. The above cited applications are herein incorporated by reference in their entirety.

FIELD

(1) Generally, the aspects of the technology described herein relate to ultrasound data collection and analysis.

BACKGROUND

(2) Ultrasound systems may be used to perform diagnostic imaging and/or treatment, using sound waves with frequencies that are higher with respect to those audible to humans. Ultrasound imaging may be used to see internal soft tissue body structures, for example to find a source of disease or to exclude any pathology. When pulses of ultrasound are transmitted into tissue (e.g., by using a pulser in an ultrasound imaging device), sound waves are reflected off the tissue, with different tissues reflecting varying degrees of sound. These reflected sound waves may then be recorded and displayed as an ultrasound image to the operator. The strength (amplitude) of the sound signal and the time it takes for the wave to travel through the body provide information used to produce the ultrasound image. Many different types of images can be formed using ultrasound systems, including real-time images. For example, images can be generated that show two-dimensional cross-sections of tissue, blood flow, motion of tissue over time, the location of blood, the presence of specific molecules, the stiffness of tissue, or the anatomy of a three-dimensional region.

SUMMARY

(3) According to one aspect, a method includes displaying, on a touch-sensitive display screen of a processing device in operative communication with an ultrasound device: an ultrasound image, a movable measurement tool, and an icon that maintains a fixed distance from a portion of the measurement tool, where the icon is configured to modify the measurement tool, and the icon does not overlap the measurement tool.

(4) In some embodiments, the measurement tool comprises a line, the icon maintains the fixed distance from an endpoint of the line, and the icon is configured to control a position of the endpoint of the line. In some embodiments, the measurement tool comprises an ellipse, the icon maintains the fixed distance from a vertex of the ellipse, and the icon is configured to control a

length of an axis of the ellipse that includes the vertex. In some embodiments, the measurement tool comprises an ellipse, the icon maintains the fixed distance from a vertex of the ellipse, and the icon is configured to control a rotation of the ellipse.

(5) According to another aspect, a method includes displaying, on a touch-sensitive display screen of a processing device in operative communication with an ultrasound device: an ultrasound image, a line extending between a first endpoint and a second endpoint, and an icon located a fixed distance from the first endpoint along a direction defined by the line; detecting a dragging movement covering a distance in a horizontal direction and/or a distance in a vertical direction across the touch-sensitive display screen, wherein the dragging movement begins on or within a threshold distance of the icon; displaying the first endpoint at a new location on the touch-sensitive display screen that is removed from the endpoint's previous location by the distance in the horizontal direction and/or the distance in the vertical direction; displaying the icon at a new location on the touch-sensitive display screen that is removed from the new location of the first endpoint by the fixed distance along the direction defined by the line; and performing a measurement on the ultrasound image based on the line.

(6) According to another aspect, a method includes displaying, on a touch-sensitive display screen of a processing device in operative communication with an ultrasound device, an ultrasound image and a line extending between a first endpoint and a second endpoint; detecting a dragging movement covering a distance in a horizontal direction and/or a distance in a vertical direction across the touch-sensitive display screen, wherein the dragging movement begins on or within a threshold distance of the line; displaying the first endpoint and the second endpoint of the line at new locations on the touch-sensitive display screen that are removed from their previous locations by the distance in the horizontal direction and/or the distance in the vertical direction; and performing a measurement on the ultrasound image based on the line.

(7) According to another aspect, a method includes displaying, on a touch-sensitive display screen of a processing device in operative communication with an ultrasound device: an ultrasound image; an ellipse having an axis that is either a major axis or a minor axis of the ellipse, wherein the axis extends between a first vertex and a second vertex of the ellipse; and an icon located a fixed distance from the first vertex along a direction defined by the axis; detecting a dragging movement covering a distance along the direction defined by the axis of the ellipse across the touch-sensitive display screen, wherein the dragging movement begins on or within a threshold distance of the icon; displaying the first vertex at a new location on the touch-sensitive display screen that is removed from the first vertex's previous location by the distance along the direction defined by the axis of the ellipse; displaying the second vertex at a new location on the touch-sensitive display screen that is removed from the second vertex's previous location by the distance along the direction defined by the axis of the ellipse; displaying the icon at a new location on the touch-sensitive display screen that is removed from the first vertex's new location by the fixed distance along the direction defined by the axis of the ellipse; and performing a measurement on the ultrasound image based on the ellipse.

(8) According to another aspect, a method includes displaying, on a touch-sensitive display screen of a processing device in operative communication with an ultrasound device: an ultrasound image; an ellipse having an axis that is either a major axis or a minor axis of the ellipse, wherein the axis extends between a first vertex and a second vertex of the ellipse; and an icon located a fixed distance from the first vertex along a direction defined by the axis; detecting a dragging movement covering a distance along and/or a distance orthogonal to the direction defined by the axis of the ellipse across the touch-sensitive display screen, wherein the dragging movement begins on or within a threshold distance of the icon; displaying the first vertex and the second vertex at new locations on the touch-sensitive display screen that are rotated from their previous locations based on the distance that is along and/or the distance orthogonal to the direction defined by the axis of the ellipse; displaying the icon at a new location on the touch-sensitive display screen that is

removed from the first vertex's new location by the fixed distance along the direction defined by the axis of the ellipse; and performing a measurement on the ultrasound image based on the ellipse.

(9) According to another aspect, a method includes displaying, on a touch-sensitive display screen of a processing device in operative communication with an ultrasound device: an ultrasound image; an ellipse having an axis that is either a major axis or a minor axis of the ellipse, wherein the axis extends between a first vertex and a second vertex of the ellipse; and an icon located a fixed distance from the first vertex along a direction defined by the axis; detecting a dragging movement covering distance in a horizontal direction and/or a distance in a vertical direction across the touch-sensitive display screen, wherein the dragging movement begins in an interior of the ellipse or within a threshold distance of a boundary of the ellipse; displaying the first vertex and the second vertex at new locations on the touch-sensitive display screen that are removed from their previous locations by the distance in the horizontal direction and/or the distance in the vertical direction; and performing a measurement on the ultrasound image based on the ellipse.

(10) According to another aspect, a method of operating a processing device configured to display ultrasound images includes displaying an ultrasound image on a display screen of the processing device; displaying a measurement tool overlay on the ultrasound image, the measurement tool overlay comprising a target point; displaying, on the display screen, a touch-sensitive measurement tool control icon corresponding to the target point; and in response to receiving touch input to the display screen, moving the target point and the touch-sensitive measurement tool control icon while maintaining a fixed distance between them. In some embodiments, the touch-sensitive measurement tool control icon does not overlap the measurement tool overlay.

(11) Some aspects include at least one non-transitory computer-readable storage medium storing processor-executable instructions that, when executed by at least one processor, cause the at least one processor to perform the above aspects and embodiments. Some aspects include an ultrasound system having a processing device configured to perform the above aspects and embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects and embodiments of the application will be described with reference to the following figures. It should be appreciated that the figures are not necessarily drawn to scale. Items appearing in multiple figures are indicated by the same reference number in all the figures in which they appear.

(2) FIG. 1 illustrates an example graphical user interface (GUI) that may be displayed on a touch-sensitive display screen of a processing device in an ultrasound system, in accordance with certain embodiments described herein. The GUI includes a line for performing a measurement on an ultrasound image;

(3) FIG. 2 illustrates another example of the graphical user interface of FIG. 1, in accordance with certain embodiments described herein;

(4) FIG. 3 illustrates another example of the graphical user interface of FIG. 1, in accordance with certain embodiments described herein;

(5) FIG. 4 illustrates another example of the graphical user interface of FIG. 1, in accordance with certain embodiments described herein;

(6) FIG. 5 illustrates another example of the graphical user interface of FIG. 1, in accordance with certain embodiments described herein;

(7) FIG. 6 illustrates an example graphical user interface that may be displayed on a touch-sensitive display screen of a processing device in an ultrasound system, in accordance with certain embodiments described herein. The GUI includes an ellipse for performing a measurement on an ultrasound image;

- (8) FIG. 7 illustrates another example of the graphical user interface of FIG. 6, in accordance with certain embodiments described herein;
- (9) FIG. 8 illustrates another example of the graphical user interface of FIG. 6, in accordance with certain embodiments described herein;
- (10) FIG. 9 illustrates another example of the graphical user interface of FIG. 6, in accordance with certain embodiments described herein;
- (11) FIG. 10 illustrates another example of the graphical user interface of FIG. 6, in accordance with certain embodiments described herein;
- (12) FIG. 11 illustrates a method for determining how much to rotate an ellipse based on a dragging movement, in accordance with certain embodiments described herein;
- (13) FIG. 12 illustrates an example GUI that may be shown when ultrasound data is being collected, in accordance with certain embodiments described herein;
- (14) FIG. 13 illustrates an example GUI that may be shown upon selection of a freeze option from the GUI of FIG. 12, in accordance with certain embodiments described herein;
- (15) FIG. 14 illustrates an example GUI that may be shown upon selection of a measurement option from FIG. 13, in accordance with certain embodiments described herein;
- (16) FIG. 15 illustrates an example process for performing measurements on an ultrasound image based on a line, in accordance with certain embodiments described herein;
- (17) FIG. 16 illustrates an example process for performing measurements on an ultrasound image based on a line, in accordance with certain embodiments described herein;
- (18) FIG. 17 illustrates an example process for performing measurements on an ultrasound image based on a line, in accordance with certain embodiments described herein;
- (19) FIG. 18 illustrates an example process for performing measurements on an ultrasound image based on a line, in accordance with certain embodiments described herein;
- (20) FIG. 19 illustrates an example process for performing measurements on an ultrasound image based on a line, in accordance with certain embodiments described herein;
- (21) FIG. 20 illustrates a schematic block diagram illustrating aspects of an example ultrasound system upon which various aspects of the technology described herein may be practiced; and
- (22) FIG. 21 is a schematic block diagram illustrating aspects of another example ultrasound system upon which various aspects of the technology described herein may be practiced.

DETAILED DESCRIPTION

(23) Conventional ultrasound systems are large, complex, and expensive systems that are typically only purchased by large medical facilities with significant financial resources. Recently, cheaper and less complex ultrasound imaging devices have been introduced. Such imaging devices may include ultrasonic transducers monolithically integrated onto a single semiconductor die to form a monolithic ultrasound device. Aspects of such ultrasound-on-a-chip devices are described in U.S. patent application Ser. No. 15/415,434 titled “UNIVERSAL ULTRASOUND DEVICE AND RELATED APPARATUS AND METHODS,” filed on Jan. 25, 2017 (and assigned to the assignee of the instant application) and published as U.S. Pat. Pub. No. US-2017-0360397-A1, which is incorporated by reference herein in its entirety. Such an ultrasound device may be in operative communication with a processing device, such as a smartphone or a tablet, having a touch-sensitive display screen. The processing device may display ultrasound images generated from ultrasound data collected by the ultrasound device.

(24) The inventors have developed technology for assisting a user in performing measurements on an ultrasound image depicted by the touch-sensitive display screen of a processing device. Performing measurements may include modifying the position, orientation, and/or shape of a measurement tool such as a line or ellipse displayed on the ultrasound image to perform calculations of spatial length or spatial area represented by the ultrasound image. The technology includes icons that are displayed a fixed distance from certain portions of a line or an ellipse, and which in some embodiments do not overlap with any portion of the line or ellipse. The icons may

be used to modify the measurement tool. For example, to modify the location of an endpoint of a line, a user may perform a dragging movement across the touch-sensitive display screen that begins on an icon located a fixed distance from the endpoint. The processing device may change the location of the endpoint by a distance corresponding to the distance covered by the dragging movement. The processing device may update, based on the dragging movement, the location of the endpoint at a sufficiently high rate such that the endpoint appears to follow the dragging movement as the dragging movement proceeds. In other words, if a user touches his/her finger to the icon and drags his/her finger across the touch-sensitive display screen, the endpoint may appear to follow the user's finger. Because changing the location of the endpoint may be initiated in this example by the user touching his/her finger to the icon, which may be located a fixed distance away from the endpoint, the endpoint may be removed from the user's finger by the fixed distance as the user drags his/her finger across the touch-sensitive display screen. Thus, as the user drags his/her finger, the endpoint may be visible to the user, and the user may be able to determine when the endpoint has moved to the desired location and release his/her finger from the touch-sensitive display to cause the endpoint to remain in the desired location. Additionally, as described above, in some embodiments the icon may not overlap with any portion of the line, which may further help the user to determine, as s/he drags his/her finger, when the line has been positioned as desired.

(25) It should be appreciated that the embodiments described herein may be implemented in any of numerous ways. Examples of specific implementations are provided below for illustrative purposes only. It should be appreciated that these embodiments and the features/capabilities provided may be used individually, all together, or in any combination of two or more, as aspects of the technology described herein are not limited in this respect.

(26) While the description below includes certain methods that a processing device may use to cause a given result to occur, a processing device may implement different methods in order to cause the same result to occur. In particular, code designed to cause the result to occur may implement a different method to cause the result to occur than those described.

(27) FIGS. 1-9 illustrate example graphical user interfaces that may be displayed on a touch-sensitive display screen of a processing device in an ultrasound system, in accordance with certain embodiments described herein. FIGS. 1-5 illustrate examples GUIs that include a line for performing a measurement on an ultrasound image. FIGS. 6-10 illustrate example GUIs that include an ellipse for performing a measurement on an ultrasound image. The processing device may be in operative communication with an ultrasound device. Ultrasound systems and devices are described in more detail with reference to FIGS. 20-21.

(28) FIG. 1 illustrates an example GUI **100** that includes a line **102**, a first icon **104**, a second icon **106**, a first crosshairs **108**, a second crosshairs **110**, a measurement value indicator **112**, a delete option **114**, and an ultrasound image **120**.

(29) The line **102** extends between a first endpoint **116** and a second endpoint **118**. The first crosshairs **108** may help to visually highlight the location of the first endpoint **116** and the second crosshairs **110** may help to visually highlight the location of the second endpoint **118**. The line **102** is superimposed on the ultrasound image **120** and may be used to perform a length measurement on the ultrasound image **120**. In particular, the processing device may perform a calculation of the spatial length represented by the ultrasound image **120** between the first endpoint **116** and the second endpoint **118**. The processing device may receive information from the ultrasound device indicating that the ultrasound image **120** was collected from an area having a certain size. The processing device may use this information to determine the spatial size represented by each pixel and thereby determine the spatial length represented by the line **102**. (Similar methods may be used for measurements of spatial length and area using an ellipse, as described below). The spatial length represented by the ultrasound image **120** between the first endpoint **116** and the second endpoint **118** is depicted by the measurement value indicator **112**. The user may cause the processing device to modify the locations of the first endpoint **116** and/or the second endpoint **118**

on the GUI **100**. For example, the user may cause the processing device to modify the locations of the first endpoint **116** and/or the second endpoint **118** to coincide with endpoints of a particular anatomical structure visible in the ultrasound image **120** if the user desires to measure the distance between the endpoints of the anatomical structure. The processing device may update the measurement value indicator **112** based on the new distance between the first endpoint **116** and the second endpoint **118**. The processing device may remove the line **102**, the first icon **104**, and the second icon **106** from the touch-sensitive display in response to a user selection of the delete option **114**.

(30) In FIG. **1**, the first icon **104** and the second icon **106** are circular, although other forms are possible. Additionally, in FIG. **1**, no portion of the first icon **104** or the second icon **106** overlaps the line **102**. However, in some embodiments, a portion of the first icon **104** or the second icon **106** may overlap the line **102**.

(31) The inventors have developed technology for assisting a user in modifying the locations of the first endpoint **116** and/or the second endpoint **118** (and thereby modifying the position and/or orientation of the line **102**) using a touch-sensitive display screen. The technology includes display of the first icon **104** and the second icon **106**. The first icon **104** is positioned a fixed distance **122** from the first endpoint **116**. The second icon **106** is positioned the fixed distance **122** from the second endpoint **118**. In some embodiments, the fixed distance **122** may be a predetermined distance. In some embodiments, the fixed distance **122** may be a default distance. In some embodiments, the fixed distance **122** may be selected by a user. In some embodiments, an icon being positioned a fixed distance from some feature (e.g., an endpoint of the line **102**) may mean that the center of the icon is positioned the fixed distance from the feature. In some embodiments, the fixed distance between the first icon **104** and the first endpoint **116** and the fixed distance between the second icon **106** and the second endpoint **118** may not be the same.

(32) The processing device may change the location of the first endpoint **116** based on a dragging movement on the touch-sensitive display screen that begins on or within a threshold distance of the first icon **104**. A dragging movement may include, for example, a user touching his/her finger to the touch-sensitive display and dragging his/her finger to a different location on the touch-sensitive display screen. The processing device may change the location of the second endpoint **118** based on a dragging movement on the touch-sensitive display screen that begins on or within a threshold distance of the second icon **106**. In particular, if a drag that begins on or within a threshold distance of the first icon **104** covers a certain distance in the horizontal direction and/or a certain distance in the vertical direction, the processing device may change the location of the first endpoint **116** by that same distance in the horizontal direction and/or distance in the vertical direction. (A drag that covers a certain distance in a certain direction need not mean that the drag actually proceeded along that direction, but rather than the drag had a component along that direction. For example, a drag in an arbitrary direction across a touch-sensitive display screen may have a component along the horizontal direction and a component along the vertical direction of the touch-sensitive display screen). If a drag that begins on or within a threshold distance of the second icon **106** covers a certain distance in the horizontal direction and/or a certain distance in the vertical direction, the processing device may change the location of the second endpoint **118** by that same distance in the horizontal direction and/or distance in the vertical direction.

(33) For example, consider the touch-sensitive display screen having an array of pixels, each pixel having a location that is x pixels in the horizontal direction and a location that is y pixels in the vertical location, where x and y are measured from an origin (e.g., a corner of the touch-sensitive display screen). Consider further that the first endpoint is located at $(e1x, e1y)$. When the user performs a dragging movement on the touch-sensitive display screen that begins at a starting location $(d1x, d1y)$ on or within a threshold distance of the first icon **104** and ends at an ending location $(d2x, d2y)$, the processing device may change the location of the first endpoint **116** such that the first endpoint **116** is displayed at $(e1x+(d2x-d1x), e1y+(d2y-d1y))$. The processing device

may similarly change the location of the second endpoint **118** based on a drag that begins at or within a threshold distance of the second icon **106**. Once the processing device has displayed the first endpoint **116** and/or the second endpoint **118** in a new location, the processing device may display the rest of the line **102** between the first endpoint **116** and/or the second endpoint **118**. In some embodiments, the processing device may use the Cartesian equation for a line to determine locations for points along the line that are not endpoints.

(34) The processing device may update, based on a dragging movement, the location of the first endpoint **116** at a sufficiently high rate such that the first endpoint **116** appears to follow the dragging movement as the dragging movement proceeds. In other words, if a user touches his/her finger to the first icon **104** and drags his/her finger across the touch-sensitive display screen, the first endpoint **116** may appear to follow the user's finger. Because changing the location of the first endpoint **116** may be initiated in this example by the user touching his/her finger to the first icon **104**, which may be located a fixed distance away from the first endpoint **116**, the first endpoint **116** may be removed from the user's finger by the fixed distance as the user drags his/her finger across the touch-sensitive display screen. Thus, as the user drags his/her finger, the first endpoint **116** may be visible to the user, and the user may be able to determine when the first endpoint **116** has moved to the desired location and release his/her finger from the touch-sensitive display to cause the first endpoint **116** to remain in the desired location. The same discussion applies to the second endpoint **118** and the second icon **106**.

(35) After a dragging movement that begins at or within a threshold distance of the first icon **104**, the processing device may change the location of the first icon **104** such that the first icon **104** is displayed a fixed distance from the first endpoint **116** along a direction defined by the line **102**. After a dragging movement that begins at or within a threshold distance of the second icon **106**, the processing device may change the location of the second icon **106** such that the second icon **106** is displayed a fixed distance from the second endpoint **118** along a direction defined by the line **102** (i.e., the direction defined by the line **102** after the location of the first endpoint **116** and/or the location of the second endpoint **118** has changed). For example, consider that after the dragging movement, the first endpoint **116** is located at $(e1x, e1y)$, the second endpoint **118** is located at $(e2x, e2y)$, and the fixed distance is d . The new location $(i1x, i1y)$ of the first icon **104** may satisfy the two equations $\sqrt{(i1x-e1x)^2+(i1y-e1y)^2}=d$ and $(i1y-e1y)/(i1x-e1x)=(c1y-c2y)/(c1x-c2x)$. It should be noted that there may be two sets of solutions for these two equations, and the solution chosen may be the one where $(i1y, i1x)$ does not coincide with the line **102**, meaning that $i1x$ is not between $e1x$ and $e2x$, and $i1y$ is not between $e1y$ and $e2y$. The processing device may similarly change the location of the second icon **106** based on a new position of the second endpoint **118**.

(36) In some embodiments, in response to a dragging movement beginning on an icon and covering a distance in the horizontal direction and/or a distance in the vertical direction, the processing device may change the location of the icon by the distance in the horizontal direction and/or the distance in the vertical direction equivalent to the distance in the horizontal direction and/or a distance in the vertical direction covered by the dragging movement, and change the location of the corresponding endpoint to be a fixed distance from the icon's new position along a direction defined by the line. In some embodiments, in response to a dragging movement beginning on an icon and covering a distance in the horizontal direction and/or a distance in the vertical direction, the processing device may change the location of both the endpoint and the icon by the distance in the horizontal direction and/or the distance in the vertical direction equivalent to the distance in the horizontal direction and/or a distance in the vertical direction covered by the dragging movement.

(37) In some embodiments, the processing device may remove the first icon **104** from display during a dragging movement that begins at the first icon **104** and remove the second icon **106** from display during a dragging movement that begins at the second icon **106**. This may help the user to understand that the measurement will be performed based on the line **102** and not based on either

the first icon **104** or the second icon **106**. In other words, this may help the user to understand that the line **102** does not extend to the first icon **104** or the second icon **106**. However, in other embodiments, the processing device may continue to display the first icon **104** and the second icon **106** during a dragging movement that begins at the first icon **104** or the second icon **106**, respectively.

(38) FIG. 2 illustrates the example graphical user interface (GUI) **100** after a dragging movement beginning on or within a threshold distance of the first icon **104**. Prior to the dragging movement, the GUI **100** may have appeared as shown in FIG. 1. The processing device has changed the location of the first endpoint **116** from its location in FIG. 1. As described above, the processing device may have changed the location of the first endpoint **116** by a distance in the horizontal direction and/or a distance in the vertical direction equivalent to the distance in the horizontal direction and/or the distance in the vertical direction covered by the dragging movement. The processing device has displayed the rest of the line **102** between the new location of the first endpoint **116** and the previous location of the second endpoint **118**. The processing device has also changed the location of the first icon **104** from its location in FIG. 1 to be the fixed distance **122** away from the first endpoint **116** along a direction defined by the line **102**. It should be noted that the processing device has changed the measurement value depicted by the measurement value indicator **112** in FIG. 2 from that shown in FIG. 1 based on the change in length of the line **102** from FIG. 1 to FIG. 2.

(39) FIG. 3 illustrates the example graphical user interface (GUI) **100** after a dragging movement beginning on or within a threshold distance of the second icon **106**. Prior to the dragging movement, the GUI **100** may have appeared as shown in FIG. 2. The processing device has changed the location of the second endpoint **118** from its location in FIG. 1. As described above, the processing device may have changed the location of the second endpoint **118** by a distance in the horizontal direction and/or a distance in the vertical direction equivalent to the distance in the horizontal direction and/or the distance in the vertical direction covered by the dragging movement. The processing device has displayed the rest of the line **102** between the new location of the second endpoint **118** and the previous location of the first endpoint **116**. The processing device has also changed the location of the second icon **106** from its location in FIG. 2 to be the fixed distance **122** away from the second endpoint **118** along a direction defined by the line **102**. It should be noted that the processing device has changed the measurement value depicted by the measurement value indicator **112** in FIG. 3 from that shown in FIG. 2 based on the change in length of the line **102** from FIG. 2 to FIG. 3.

(40) In some embodiments, the processing device may change the position of both the first endpoint **116** and the second endpoint **118** based on a dragging movement that begins on or within a threshold distance of any portion of the line **102**. When the user performs a dragging movement on the touch-sensitive display screen that begins at a starting location ($d1x, d1y$) on or within a threshold distance of the line **102** and ends at an ending location ($d2x, d2y$), the processing device may change the locations of both the first endpoint **116** and the second endpoint **118** by a distance of $(d2x-d1x, d2y-d1y)$. The processing device may also change the locations of the first icon **104** and the second icon **106** such that they are the fixed distance **122** away from the first endpoint **116** and the second endpoint **118**, respectively, along the direction of the line **102**. Once the processing device has displayed the first endpoint **116** and the second endpoint **118** in new locations, the processing device may display the rest of the line **102** between the new locations of the first endpoint **116** and the second endpoint **118**. In some embodiments, the processing device may use the Cartesian equation for a line to determine locations for points along the line **102** between the first endpoint **116** and the second endpoint **118**. In some embodiments, the processing device may change the locations of all displayed points along the line **102** by a distance of $(d2x-d1x, d2y-d1y)$.

(41) FIG. 4 illustrates the example graphical user interface (GUI) **100** after a dragging movement beginning on or within a threshold distance of the line **102**. Prior to the dragging movement, the

GUI **100** may have appeared as shown in FIG. **3**. The processing device has changed the location of the line from its location in FIG. **3**. As described above, the processing device may have changed the location of the first endpoint **116** and the second endpoint **118** by a distance in the horizontal direction and/or a distance in the vertical direction equivalent to the distance in the horizontal direction and/or the distance in the vertical direction covered by the dragging movement. The processing device has displayed the rest of the line **102** between the new locations of the first endpoint **116** and the second endpoint **118**. The processing device has also changed the locations of the first icon **104** and the second icon **106** from their locations in FIG. **3** to be the fixed distance **122** away from the first endpoint **116** and the second endpoint **116**, respectively, along a direction defined by the line **102**.

(42) FIG. **5** illustrates the example graphical user interface (GUI) **100** during a dragging movement beginning on or within a threshold distance of the first icon **104**. The GUI **100** in FIG. **5** is similar to that shown in FIG. **2**, with the addition of an inset **524**. The inset **524** depicts a magnification of a portion **526** of the ultrasound image **120**. In particular, the inset **524** depicts a portion **526** of the ultrasound image **120** that is proximal to the first endpoint **116**. The inset **524** further depicts the first endpoint **116**, the first crosshairs **108**, and a portion of the line **102** that is within the portion **526** of the ultrasound image **120**. The processing device may display the inset **524** when the user begins a dragging movement and continue to display the inset **524** as the user continues the dragging movement. Because the inset **524** illustrates the magnified portion **526** of the ultrasound image **120** that is proximal to the first endpoint **116**, the user may use the inset **524** to determine how to perform the dragging movement in order to change the location of the first endpoint **116** to the desired location on the ultrasound image **120**, and also to determine when the first endpoint **116** is at the desired location. If the user begins a dragging movement on or within a threshold distance of the second icon **106**, the processing device may display the inset **524** and show a magnified portion (not shown in FIG. **5**) of the ultrasound image **120** that is proximal to the second endpoint **118**, and the inset **524** may also depict the second endpoint **118**, the second crosshairs **110**, and a portion of the line **102** that is within the portion of the ultrasound image **120** depicted by the inset **524**. It should be noted that in FIG. **5**, in contrast to FIG. **2**, the processing device does not display the first icon **104** during the dragging movement that began on or within a threshold distance of the first icon **104**. However, in some embodiments, the processing may display the first icon **104** during the dragging movement. In some embodiments, the processing device may not display the inset **524** during a dragging movement.

(43) It should be understood that in some embodiments, certain portions of the GUI **100** may be absent. For example, the first crosshairs **108**, the second crosshairs **110**, and/or the delete option **114** may be absent. In some embodiments, the measurement value indicator **112** may have a different form than shown and/or be located at a different location on the touch-sensitive display screen. Additionally, while the GUI **100** shows certain other features that are not described herein (e.g., certain buttons or indicators), in some embodiments such features may be absent or different.

(44) FIG. **6** illustrates an example graphical user interface (GUI) **600** that includes an ellipse **638**, a first icon **640**, a second icon **662**, a first measurement value indicator **656**, a second measurement value indicator **658**, a delete option **660**, and an ultrasound image **120**. The second icon **662** includes a first arrow **652** and a second arrow **654**.

(45) The ellipse **638** includes a center location **664**, a first axis **674**, and a second axis **676**. The first axis **674** extends between two endpoints, namely a first vertex **642** and a second vertex **644** of the ellipse **638**. The second axis **676** extends between two endpoints, namely a third vertex **646** and a fourth vertex **648** of the ellipse **638**. The first axis **674** and the second axis **676** may be equivalent to the major axis and the minor axis of the ellipse, or vice versa. It should be appreciated that the ellipse **638** may be a circle. The ellipse **638** is superimposed on the ultrasound image **120** and may be used by the processing device to perform a measurement on the ultrasound image **120**. In FIG. **6**, the processing device displays the value of the spatial length represented by the ultrasound image

120 along the circumference of the ellipse **638** with the first measurement value indicator **656** and the processing device displays the value of the spatial area represented by the ultrasound image **120** within the ellipse **638** with the second measurement value indicator **658**. The user may cause the processing device to modify the ellipse (e.g., the position, orientation, and/or shape of the ellipse). For example, the user may cause the processing device to modify the ellipse to coincide with a particular anatomical structure visible in the ultrasound image **120** if the user desires to measure the circumference or area of the anatomical structure as depicted by the ultrasound image **120**.

(46) The inventors have developed technology for assisting a user in modifying the position, orientation, and shape of the ellipse **638** using a touch-sensitive display screen. The technology includes display of the first icon **640** and the second icon **662**. The processing device displays the first icon **640** a fixed distance **650** from the first vertex **642**. The processing device displays the second icon **662** the fixed distance **650** from the fourth vertex **648**. In some embodiments, the fixed distance **650** may be a predetermined distance. In some embodiments, the fixed distance **650** may be a default distance. In some embodiments, the fixed distance **650** may be selected by a user. In some embodiments, an icon being positioned a fixed distance from some feature (e.g., a vertex of the ellipse **638**) may mean that the center of the icon is positioned the fixed distance from the feature. In some embodiments, the fixed distance between the first icon **640** and the first vertex **642** and the fixed distance between the second icon **662** and the third vertex **646** may not be the same.

(47) In FIG. **6**, the first icon **640** and the second icon **662** are circular, although other forms are possible. Additionally, in FIG. **6**, no portion of the first icon **640** or the second icon **662** overlaps the ellipse **638**. However, in some embodiments, a portion of the first icon **640** or the second icon **662** may overlap the ellipse **638**.

(48) The processing device may change the length of the first axis **674** based on a dragging movement on the touch-sensitive display screen that begins on or within a threshold distance of the first icon **640**. In particular, if the drag that begins on or within a threshold distance of the first icon **640** covers a certain distance away from the ellipse **638** along the direction defined by the first axis **674**, the processing device may change the locations of the first vertex **642** and the second vertex **644** by that same distance away from the ellipse along the direction defined by the first axis **674**. In other words, the processing device may expand the first axis **674** of the ellipse **638** by two times the distance along the direction defined by the first axis **674**. If the drag that begins on or within a threshold distance of the first icon **640** covers a certain distance towards from the ellipse **638** along the direction defined by the first axis **674**, the processing device may change the locations of the first vertex **642** and the second vertex **644** by that same distance towards from the ellipse along the direction defined by the first axis **674**. In other words, the processing device may contract the first axis **674** of the ellipse **638** by two times the distance along the direction defined by the first axis **674**. The processing device may similarly change the length of the second axis **676** based on a dragging movement on the touch-sensitive display screen that begins on or within a threshold distance of the second icon **662**. The processing device may display other points along the ellipse **638** based on new lengths of the first axis **674** and/or the second axis **676**. For example, the processing device may determine new locations for other points along the ellipse **638** based on the Cartesian equation for an ellipse **638**. In some embodiments, to display the ellipse **638** based on the Cartesian equation for an ellipse **638**, the processing device may only use the center location **664** of the ellipse, one of the first vertex **642** and the second vertex **644**, and one of the third vertex **646** and the fourth vertex **648**.

(49) Consider the touch-sensitive display screen having an array of pixels, each pixel having a location that is x pixels in the horizontal direction and a location that is y pixels in the vertical location, where x and y are measured from an origin (e.g., a corner of the touch-sensitive display screen). For simplicity, assume the first axis **674** of the ellipse **638** is parallel to the vertical direction of the touch-sensitive display and the second axis **676** of the ellipse is parallel to the horizontal direction of the touch-sensitive display. Consider further that the first vertex **642** is

located at (v1x, v1y), the second vertex **644** is located at (v2x, v2y), and the first icon is located at (i1x, i1y). When the user performs a dragging movement on the touch-sensitive display screen that begins at a starting location (d1x, d1y) on or within a threshold distance of the first icon **640** and ends at an ending location (d2x, d2y), the processing device may change the location of the first vertex **642** such that the first vertex **642** is displayed at (v1x, v1y+(d2y-d1y)). The processing device may also change the location of the second vertex **644** such that the second vertex **644** is displayed at (v2x, v2y-(d2y-d1y)). As described above, the processing device may display other points along the ellipse **638** based on the new locations of the first vertex **642** and the second vertex **644**.

(50) The processing device may update, based on a dragging movement, the location of the first vertex **642** at a sufficiently high rate such that the first vertex **642** appears to follow the dragging movement as the dragging movement proceeds. In other words, if a user touches his/her finger to the first icon **640** and drags his/her finger across the touch-sensitive display screen, the first vertex **642** may appear to follow the user's finger. Because changing the location of the first vertex **642** may be initiated in this example by the user touching his/her finger to the first icon **640**, which may be located a fixed distance away from the first vertex **642**, the first vertex **642** may be removed from the user's finger by the fixed distance as the user drags his/her finger across the touch-sensitive display screen. Thus, as the user drags his/her finger, the first vertex **642** may be visible to the user, and the user may be able to determine when the first vertex **642** has moved to the desired location and release his/her finger from the touch-sensitive display to cause the first vertex **642** to remain in the desired location. The same discussion applies to the fourth vertex **648** and the second icon **662**.

(51) After a dragging movement, the processing device may change the location of the first icon **640** such that the first icon **640** is displayed a fixed distance from the first vertex **642** along the direction defined by the first axis **674**. For example, consider that after the dragging movement, the first vertex **642** is located at (v1x, v1y), the second vertex **644** is located at (v2x, v2y), and the fixed distance is d. The new location (i1x, i1y) of the first icon **640** may satisfy the two equations $\sqrt{(i1x-v1x)^2+(i1y-v1y)^2}=d$ and $(i1y-v1y)/(i1x-v1x)=(v1y-v2y)/(v1x-v2x)$. It should be noted that there may be two sets of solutions for these two equations, and the solution chosen may be the one where (i1y, i1x) is not within the ellipse **630**, meaning that i1x is not between v1x and v2x, and i1y is not between v1y and v2y.

(52) In the simplified example described above, the direction defined by the first axis **674** is along the vertical direction of the touch-sensitive display, but in the general case where the direction defined by the first axis **674** is rotated to an angle relative to the vertical direction of the touch-sensitive display, the expressions above may be modified to account for such rotation. In a similar manner as described above regarding changing the location of the first vertex **642**, the second vertex **644**, and the first icon **640**, the processing device may change the location of the third vertex **646**, the fourth vertex **648**, and the second icon **662** based on a dragging movement that begins on or within a threshold distance of the second icon **662** and covers a certain distance away from/toward the ellipse **638** along the direction defined by the second axis **676**.

(53) FIG. 7 illustrates the example graphical user interface (GUI) **600** after a dragging movement beginning on or within a threshold distance of the first icon **640** and covering a distance towards the ellipse **638** along the direction of the first axis **674**. Prior to the dragging movement, the GUI **600** may have appeared as shown in FIG. 6. The processing device has changed the locations of the first vertex **642** and the second vertex **644** from their locations in FIG. 6. (In other words, the processing device has changed the length of the first axis **674**.) As described above, the processing device may have changed the locations of the first vertex **642** and the second vertex **644** by the distance covered by the dragging movement along the direction defined by the first axis **674**. The processing device has also changed the location of the first icon **640** from its location in FIG. 6 to

be the fixed distance **650** away from the first vertex **674** along a direction defined by the first axis **674**. It should be noted that the processing device has changed the measurement values depicted by the first measurement value indicator **656** and the second measurement value indication **658** from that shown in FIG. **6** based on the change in length of the first axis **674** from FIG. **6** to FIG. **7**.

(54) FIG. **8** illustrates the example graphical user interface (GUI) **600** after a dragging movement beginning on or within a threshold distance of the second icon **662** and covering a distance towards the ellipse **638** along the direction of the second axis **676**. Prior to the dragging movement, the GUI **600** may have appeared as shown in FIG. **7**. The processing device has changed the locations of the third vertex **646** and the fourth vertex **648** from their locations in FIG. **7**. (In other words, the processing device has changed the length of the second axis **676**.) As described above, the processing device may have changed the locations of the third vertex **646** and the fourth vertex **648** by the distance covered by the dragging movement along the direction defined by the second axis **676**. The processing device has also changed the location of the second icon **662** from its location in FIG. **7** to be the fixed distance **650** away from the fourth vertex **648** along a direction defined by the second axis **676**. It should be noted that the processing device has changed the measurement values depicted by the first measurement value indicator **656** and the second measurement value indicator **658** from that shown in FIG. **7** based on the change in length of the second axis **676** from FIG. **7** to FIG. **8**.

(55) In some embodiments, the processing device may change the position of the ellipse **638** based on a dragging movement that begins in the interior of the ellipse **638**, on the boundary of the ellipse **638**, or within a threshold distance of the boundary of the ellipse **638**. When the user performs a dragging movement on the touch-sensitive display screen that begins at a starting location ($d1x$, $d1y$) in the interior of the ellipse **638** or within a threshold distance of the boundary of the ellipse **638** and ends at an ending location ($d2x$, $d2y$), the processing device may change the locations of every point on the ellipse **638**, as well as the first icon **640** and the second icon **662**, by a distance of ($d2x-d1x$, $d2y-d1y$). In some embodiments, rather than moving every point on the ellipse **638** by a specific distance, the processing device may change the locations of the center **664** of the ellipse **638**, the first vertex **642**, the second vertex **644**, the third vertex **646**, and the fourth vertex **648** by the specific distance and display the rest of the ellipse **638** based on these new locations using the Cartesian equation for an ellipse **638**.

(56) FIG. **9** illustrates the example graphical user interface (GUI) **600** after a dragging movement beginning in the interior of the ellipse **638** or within a threshold distance of the boundary of the ellipse **638**. Prior to the dragging movement, the GUI **600** may have appeared as shown in FIG. **8**. The processing device has changed the position (but not the orientation or shape) of the ellipse **638** by the distance covered by the dragging movement. The processing device has also changed the location of the first icon **640** from its location in FIG. **8** to be the fixed distance **650** away from the first vertex **674** along a direction defined by the first axis **674**, and the location of the second icon **662** from its location in FIG. **8** to be the fixed distance **650** away from the fourth vertex **648** along a direction defined by the second axis **676**.

(57) In some embodiments, the processing device may rotate the ellipse **638** based on a dragging movement that begins on or at the second icon **662** and covers a distance along and/or a distance orthogonal to the direction of the second axis **676** of the ellipse **638**. FIG. **10** illustrates the example graphical user interface (GUI) **600** after a dragging movement beginning on or within a threshold distance of the second icon **662** and covering a distance along and/or a distance orthogonal to the direction of the second axis **676**. Prior to the dragging movement, the GUI **600** may have appeared as shown in FIG. **9**. As described above, the processing device has rotated the locations of every point of the ellipse **638** based on the drag distance along and/or the drag distance orthogonal to the direction of the second axis **676**. The processing device has also changed the location of the first icon **640** from its location in FIG. **4** to be the fixed distance **650** away from the first vertex **674** along a direction defined by the first axis **674**, and the location of the second icon **662** from its

location in FIG. 8 to be the fixed distance 650 away from the fourth vertex 648 along a direction defined by the second axis 676.

(58) FIG. 11 illustrates a method for determining how much to rotate the ellipse 638 based on the dragging movement, in accordance with certain embodiments described herein. The left side of FIG. 11 shows the ellipse 638 before the dragging movement and the right side of FIG. 11 shows the ellipse 638 after the dragging movement. For simplicity, before the dragging movement, the center location 664 is labeled C, the second vertex 644 is labeled A, and the first vertex 642 is labeled B. After the dragging movement, the center location 664 is labeled C', the location of the second vertex 644 is labeled A', the location of the first vertex 642 is labeled B', and the location of the second icon 662 is labeled D'. The dragging movement begins at the location of the second icon 662 and ends at a location separated from the previous location by a vector V (where V may have components along and/or orthogonal to the second axis 676). The processing device may determine the location of C' to be the same as C, namely, the center location 664 may not change. The processing device may determine the location of A' to be A+V, in other words, the previous location of the second vertex 644 plus the vector of the dragging movement. The processing device may determine the location of B' to be C+normal(A'C)*length(BC). In other words, the new location of the first vertex 642 may be the center location 664 plus a vector that has a length equal to the distance between the center location 664 and the previous location of the first vertex 642, and a direction that is perpendicular to a vector between the center location 664 and the new location of the second vertex 644. The processing device may determine new locations for the rest of the points on the ellipse 638 based on the new locations for the first vertex 642 and the second vertex 644.

(59) It should be appreciated from the above description of FIG. 11 that rotations of the ellipse 638 may be controlled both by components of a drag distance beginning at or within a threshold distance of the second icon 662 (in other words, the components of the vector V) that are along the direction of the second axis 676 and orthogonal to the direction of the second axis 676. As described above, a dragging movement beginning at or within a threshold distance of the second icon 662 and covering a distance along the direction of the second axis 676 may also control the length of the second axis 676. Thus, a dragging movement beginning at or within a threshold distance of the second icon 662 and having only a component along the direction of the second axis 676 may only modify the length of the second axis 676. A dragging movement beginning at or within a threshold distance of the second icon 662 and having components both along and orthogonal to the direction of the second axis 676 may modify both the length of the second axis 676 and the rotation of the ellipse 638. The description of FIG. 11 may apply both to the general case of a dragging movement having components both along and orthogonal to the direction of the second axis 676, as well as the special case of a dragging movement having a component only along the direction of the second axis 676. In some embodiments, the processing device may use a different method for determining how to rotate an ellipse than the method illustrated by FIG. 11.

(60) The first arrow 652 and the second arrow 654 may serve to indicate to a user that the second icon 662 (as opposed to the first icon 640) can be used to rotate the ellipse 638. In some embodiments, the positioning of the first arrow 652 and the second arrow 654 may change as the shape of the ellipse 638 changes so that the arrows approximate the curvature of the ellipse 638.

(61) It should be understood that in some embodiments, certain portions of the GUI 600 may be absent. For example, the second arrow 654, the first arrow 652, the first measurement value indicator 656, the second measurement value indicator 658, and/or the delete option 660 may be absent. In some embodiments, the first measurement value indicator 656 and/or the second measurement value indicator 658 may have different forms than shown and/or be located at a different locations on the touch-sensitive display screen. Additionally, while the GUI 600 shows certain other features that are not described herein (e.g., certain buttons or indicators), in some embodiments such features may be absent or different.

(62) While the above description has described that a processing device may perform certain calculations using pixels, in some embodiments the processing device may perform calculations using points. It should be noted that certain calculations described herein may produce fractional pixel results. In some embodiments, fractional pixel results may be rounded to a whole pixel. In some embodiments, the processing device may use antialiasing to interpret pixel values for a fractional pixel result (e.g., to interpret pixel values for pixels (1, 1) and (2, 1) when a calculation indicates that something should be displayed at pixel (1.5, 1)). As described above, the processing device may change the location of one feature of a measurement tool (e.g., a line or an ellipse) based on a dragging movement that begins on or within a threshold distance of a certain feature. In some embodiments, the distance may be measured in pixels (e.g., 30 pixels). While the above description has described a touch-sensitive display screen, in some embodiments the screen may not be touch-sensitive display screen, and a click and drag of a cursor (e.g., using a mouse) may be the equivalent of a dragging movement.

(63) FIG. 12 illustrates an example GUI 1200 that may be shown when ultrasound data is being collected, in accordance with certain embodiments described herein. The GUI 1200 depicts the most recent ultrasound image 120 collected by the processing device from the ultrasound device. As further ultrasound images 120 are collected, the processing device may continuously update the GUI 1200 to depict the most recent ultrasound image 120 collected. The GUI 1200 further includes a freeze option 1226.

(64) FIG. 13 illustrates an example GUI 1300 that may be shown upon selection of the freeze option 1226, in accordance with certain embodiments described herein. The GUI 1300 depicts the most recent ultrasound image 120 collected by the processing device from the ultrasound device when the freeze option 1226 was selected. In other words, the processing device freezes the most recent ultrasound 120 on the GUI 1300, and the processing device may not update the GUI 1300 with ultrasound images 120 that are collected subsequently. In the GUI 1300, the freeze option 1226 can have a different color or pattern, which may indicate that the GUI 1300 is currently showing a frozen ultrasound image 120. Additionally, the GUI 1300 depicts a measurement option 1328.

(65) FIG. 14 illustrates an example GUI 1400 that may be shown upon selection of the measurement option 1328, in accordance with certain embodiments described herein. The GUI 1400 can depict the freeze option 1226 having a different color or pattern as explained with respect to FIG. 13, as well as a label option 1430, an ellipse measurement option 1432, a line measurement option 1434, and a menu close option 1436. Upon selection of the label option 1430, the processing device may display a GUI enabling a user to place labels on the ultrasound image 130. Upon selection of the ellipse measurement option 1432, the processing device may display the GUI 600, with the ellipse 638, the first icon 640, and the second icon 662 shown in default positions. Upon selection of the line measurement option 1434, the processing device may display the GUI 100, with the line 102, the first icon 104, and the second icon 106 shown in default positions. Upon selection of the menu close option 1436, the processing device may display the GUI 1300 (i.e., remove from display the label option 1430, the ellipse measurement option 1432, and the line measurement option 1434).

(66) FIGS. 15-19 illustrate example processes for performing measurements on an ultrasound image, in accordance with certain embodiments described herein. The processes may be performed by a processing device in an ultrasound system. The processing device may be, for example, a mobile phone, tablet, or laptop in operative communication with an ultrasound probe. The ultrasound probe and the processing device may communicate over a wired communication link (e.g., over Ethernet, a Universal Serial Bus (USB) cable or a Lightning cable) or over a wireless communication link (e.g., over a BLUETOOTH, WiFi, or ZIGBEE wireless communication link).

(67) FIG. 15 illustrates an example process 1500 for performing measurements on an ultrasound image based on a line, in accordance with certain embodiments described herein. Further

description of the process **1500** may be found with reference to FIGS. **1-5**.

(68) In act **1502**, the processing device displays, on a touch-sensitive display screen, (1) an ultrasound image, (2) a line extending between a first endpoint and a second endpoint and (3) an icon located a fixed distance from the first endpoint along a direction defined by the line. The process **1500** proceeds from act **1502** to act **1504**.

(69) In act **1504**, the processing device detects a dragging movement covering a distance in the horizontal direction and/or a distance in the vertical direction across the touch-sensitive display screen, where the dragging movement begins on or within a threshold distance of the icon. The process **1500** proceeds from act **1504** to act **1506**.

(70) In act **1506**, the processing device displays the first endpoint at a new location on the touch-sensitive display screen that is removed from the endpoint's previous location by the distance in the horizontal direction and/or the distance in the vertical direction covered by the dragging movement. The process **1500** proceeds from act **1506** to act **1508**.

(71) In act **1508**, the processing device displays the icon at a new location on the touch-sensitive display screen that is removed from the new location of the first endpoint by the fixed distance along the direction defined by the line. The process **1500** proceeds from act **1508** to act **1510**.

(72) In act **1510**, the processing device performs a measurement on the ultrasound image based on the line. For example, the processing device may perform a calculation of the spatial length represented by the ultrasound image between the first endpoint and the second endpoint of the line. In some embodiments, the processing device may display the result of the measurement.

(73) In some embodiments, certain acts of the process **1500** may be absent. For example, in some embodiments, act **1510** may be absent. In some embodiments, acts **1504-1510** may be absent. In some embodiments, acts **1504-1508** may be absent. In some embodiments, other combinations of acts may be absent.

(74) FIG. **16** illustrates an example process **1600** for performing measurements on an ultrasound image based on a line, in accordance with certain embodiments described herein. Further description of the process **1600** may be found with reference to FIG. **4**.

(75) In act **1602**, the processing device displays, on a touch-sensitive display screen, (1) an ultrasound image and (2) a line extending between a first endpoint and a second endpoint. The process **1600** proceeds from act **1602** to act **1604**.

(76) In act **1604**, the processing device detects a dragging movement covering a distance in the horizontal direction and/or a distance in the vertical direction across the touch-sensitive display screen, where the dragging movement begins on or within a threshold distance of the line. The process **1600** proceeds from act **1604** to act **1606**.

(77) In act **1606**, the processing device displays both the first endpoint and the second endpoint of the line at new locations on the touch-sensitive display screen that are removed from their previous locations by the distance in the horizontal direction and/or the distance in the vertical direction. The process **1600** proceeds from act **1606** to act **1608**.

(78) In act **1608**, the processing device performs a measurement on the ultrasound image based on the line. For example, the processing device may perform a calculation of the spatial length represented by the ultrasound image between the first endpoint and the second endpoint of the line. In some embodiments, the processing device may display the result of the measurement.

(79) In some embodiments, certain acts of the process **1600** may be absent. For example, in some embodiments, act **1608** may be absent. In some embodiments, acts **1604-1608** may be absent. In some embodiments, acts **1604-1606** may be absent. In some embodiments, other combinations of acts may be absent.

(80) FIG. **17** illustrates an example process **1700** for performing measurements on an ultrasound image based on an ellipse, in accordance with certain embodiments described herein. Further description of the process **1700** may be found with reference to FIGS. **6-8**.

(81) In act **1702**, the processing device displays, on a touch-sensitive display screen, (1) an

ultrasound image, (2) an ellipse having an axis that is either the major or minor axis of the ellipse, where the axis extends between a first vertex and a second vertex; and (3) an icon located a fixed distance from the first vertex along a direction defined by the axis. The process **1700** proceeds from act **1702** to act **1704**.

(82) In act **1704**, the processing device detects a dragging movement covering a distance along the direction defined by the axis of the ellipse across the touch-sensitive display screen, where the dragging movement begins on or within a threshold distance of the icon. The process **1700** proceeds from act **1704** to act **1706**.

(83) In act **1706**, the processing device displays the first vertex at a new location on the touch-sensitive display screen that is removed from the first vertex's previous location by the distance along the direction defined by the axis of the ellipse covered by the dragging movement. The process **1700** proceeds from act **1706** to act **1708**.

(84) In act **1708**, the processing device displays the second vertex at a new location on the touch-sensitive display screen that is removed from the second vertex's previous location by the distance along the direction defined by the axis of the ellipse covered by the dragging movement. The process **1700** proceeds from act **1708** to act **1710**.

(85) In act **1710**, the processing device displays the icon at a new location on the touch-sensitive display screen that is removed from the first vertex's new location by the fixed distance along the direction defined by the axis of the ellipse. The process **1700** proceeds from act **1710** to act **1712**.

(86) In act **1712**, the processing device performs a measurement on the ultrasound image based on the ellipse. For example, the processing device may perform a calculation of the spatial length represented by the ultrasound image along the circumference of the ellipse or a calculation of the spatial area represented by the ultrasound image within the ellipse. In some embodiments, the processing device may display the result of the measurement.

(87) In some embodiments, certain acts of the process **1700** may be absent. For example, in some embodiments, act **1712** may be absent. In some embodiments, acts **1704-1712** may be absent. In some embodiments, acts **1704-1710** may be absent. In some embodiments, other combinations of acts may be absent.

(88) FIG. **18** illustrates an example process **1800** for performing measurements on an ultrasound image based on an ellipse, in accordance with certain embodiments described herein. Further description of the process **1800** may be found with reference to FIGS. **10-11**.

(89) In act **1802**, the processing device displays, on a touch-sensitive display screen, (1) an ultrasound image, (2) an ellipse having an axis that is either the major or minor axis of the ellipse, where the axis extends between a first vertex and a second vertex; and (3) an icon located a fixed distance from the first vertex along a direction defined by the axis. The process **1800** proceeds from act **1802** to act **1804**.

(90) In act **1804**, the processing device detects a dragging movement covering a distance along and/or a distance orthogonal to the direction defined by the axis of the ellipse across the touch-sensitive display screen, where the dragging movement begins on or within a threshold distance of the icon. The process **1800** proceeds from act **1804** to act **1806**.

(91) In act **1806**, the processing device displays the first vertex and the second vertex at new locations on the touch-sensitive display screen that are rotated from their previous locations based on the distance along and/or the distance orthogonal to the direction defined by the axis of the ellipse that is covered by the dragging movement. The process **1800** proceeds from act **1806** to act **1808**.

(92) In act **1808**, the processing device displays the icon at a new location on the touch-sensitive display screen that is removed from the first vertex's new location by the fixed distance along the direction defined by the axis of the ellipse. The process **1800** proceeds from act **1808** to act **1810**.

(93) In act **1810**, the processing device performs a measurement on the ultrasound image based on the ellipse. For example, the processing device may perform a calculation of the spatial length

represented by the ultrasound image along the circumference of the ellipse or a calculation of the spatial area represented by the ultrasound image within the ellipse. In some embodiments, the processing device may display the result of the measurement.

(94) In some embodiments, certain acts of the process **1800** may be absent. For example, in some embodiments, act **1810** may be absent. In some embodiments, acts **1804-1810** may be absent. In some embodiments, acts **1804-1808** may be absent. In some embodiments, other combinations of acts may be absent.

(95) FIG. **19** illustrates an example process **1900** for performing measurements on an ultrasound image based on an ellipse, in accordance with certain embodiments described herein. Further description of the process **1900** may be found with reference to FIG. **9**.

(96) In act **1902**, the processing device displays, on a touch-sensitive display screen, (1) an ultrasound image, (2) an ellipse having an axis that is either the major or minor axis of the ellipse, wherein the axis extends between a first vertex and a second vertex; and (3) an icon located a fixed distance from the first vertex along a direction defined by the axis. The process **1900** proceeds from act **1902** to act **1904**.

(97) In act **1904**, the processing device detects a dragging movement covering distance in the horizontal direction and/or a distance in the vertical direction across the touch-sensitive display screen, where the dragging movement begins in the interior of the ellipse or within a threshold distance of the boundary of the ellipse. The process **1900** proceeds from act **1904** to act **1906**.

(98) In act **1906**, the processing device displays the first vertex and the second vertex at new locations on the touch-sensitive display screen that are removed from their previous locations by the distance in the horizontal direction and/or the distance in the vertical direction covered by the dragging movement. The process **1900** proceeds from act **1906** to act **1908**.

(99) In act **1908**, the processing device performs a measurement performed on the ultrasound image based on the ellipse. For example, the processing device may perform a calculation of the spatial length represented by the ultrasound image along the circumference of the ellipse or a calculation of the spatial area represented by the ultrasound image within the ellipse. In some embodiments, the processing device may display the result of the measurement.

(100) In some embodiments, certain acts of the process **1900** may be absent. For example, in some embodiments, act **1910** may be absent. In some embodiments, acts **1904-1910** may be absent. In some embodiments, acts **1904-1908** may be absent. In some embodiments, other combinations of acts may be absent.

(101) The above description has described that a user may modify measurement tools (e.g., a line or an ellipse) through dragging movements that begin on or within a threshold distance of an icon that is located a fixed distance from a portion of the measurement tool. In some embodiments, one or more of the icons described above may be absent, and a user may modify measurement tools through dragging movements that begin on or within a threshold distance of a region of the touch-sensitive display screen that is located the fixed distance from the portion of the measurement tool, even though the region does not contain an icon.

(102) The above description has described modifying measurement tools (e.g., a line or an ellipse) based on a distance in the horizontal and/or vertical direction covered by a dragging movement. In some embodiments, the processing device may modify measurement tools based on taps. In particular, a user may tap an icon and then another location on the touch-sensitive display screen. The processing device may then modify the measurement tool based on the distance in the horizontal and/or vertical direction between the two tapped locations.

(103) Various inventive concepts may be embodied as one or more processes, of which examples have been provided. The acts performed as part of each process may be ordered in any suitable way. Thus, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments. Further, one or more of the processes may be combined

and/or omitted, and one or more of the processes may include additional steps.

(104) FIG. **20** illustrates a schematic block diagram illustrating aspects of an example ultrasound system **2000** upon which various aspects of the technology described herein may be practiced. For example, one or more components of the ultrasound system **2000** may perform any of the processes described herein. As shown, the ultrasound system **2000** includes processing circuitry **2001**, input/output devices **2003**, ultrasound circuitry **2005**, and memory circuitry **2007**.

(105) The ultrasound circuitry **2005** may be configured to generate ultrasound data that may be employed to generate an ultrasound image. The ultrasound circuitry **2005** may include one or more ultrasonic transducers monolithically integrated onto a single semiconductor die. The ultrasonic transducers may include, for example, one or more capacitive micromachined ultrasonic transducers (CMUTs), one or more CMOS ultrasonic transducers (CUTs), one or more piezoelectric micromachined ultrasonic transducers (PMUTs), and/or one or more other suitable ultrasonic transducer cells. In some embodiments, the ultrasonic transducers may be formed the same chip as other electronic components in the ultrasound circuitry **2005** (e.g., transmit circuitry, receive circuitry, control circuitry, power management circuitry, and processing circuitry) to form a monolithic ultrasound imaging device.

(106) The processing circuitry **2001** may be configured to perform any of the functionality described herein. The processing circuitry **2001** may include one or more processors (e.g., computer hardware processors). To perform one or more functions, the processing circuitry **2001** may execute one or more processor-executable instructions stored in the memory circuitry **2007**. The memory circuitry **2007** may be used for storing programs and data during operation of the ultrasound system **2000**. The memory circuitry **2007** may include one or more storage devices such as non-transitory computer-readable storage media. The processing circuitry **2001** may control writing data to and reading data from the memory circuitry **2007** in any suitable manner.

(107) In some embodiments, the processing circuitry **2001** may include specially-programmed and/or special-purpose hardware such as an application-specific integrated circuit (ASIC). For example, the processing circuitry **2001** may include one or more graphics processing units (GPUs) and/or one or more tensor processing units (TPUs). TPUs may be ASICs specifically designed for machine learning (e.g., deep learning). The TPUs may be employed to, for example, accelerate the inference phase of a neural network.

(108) The input/output (I/O) devices **2003** may be configured to facilitate communication with other systems and/or an operator. Example I/O devices **2003** that may facilitate communication with an operator include: a keyboard, a mouse, a trackball, a microphone, a touch-sensitive display screen, a printing device, a display screen, a speaker, and a vibration device. Example I/O devices **2003** that may facilitate communication with other systems include wired and/or wireless communication circuitry such as BLUETOOTH, ZIGBEE, Ethernet, WiFi, and/or USB communication circuitry.

(109) It should be appreciated that the ultrasound system **2000** may be implemented using any number of devices. For example, the components of the ultrasound system **2000** may be integrated into a single device. In another example, the ultrasound circuitry **2005** may be integrated into an ultrasound imaging device that is communicatively coupled with a processing device that includes the processing circuitry **2001**, the input/output devices **2003**, and the memory circuitry **2007**.

(110) FIG. **21** is a schematic block diagram illustrating aspects of another example ultrasound system **2100** upon which various aspects of the technology described herein may be practiced. For example, one or more components of the ultrasound system **2100** may perform any of the processes described herein. As shown, the ultrasound system **2100** includes an ultrasound imaging device **2114** in wired and/or wireless communication with a processing device **2102**. The processing device **2102** includes an audio output device **2104**, an imaging device **2106**, a display screen **2108**, a processor **2110**, a memory **2112**, and a vibration device **2109**. The processing device **2102** may communicate with one or more external devices over a network **2116**. For example, the processing

device **2102** may communicate with one or more workstations **2120**, servers **2118**, and/or databases **2122**.

(111) The ultrasound imaging device **2114** may be configured to generate ultrasound data that may be employed to generate an ultrasound image. The ultrasound imaging device **2114** may be constructed in any of a variety of ways. In some embodiments, the ultrasound imaging device **2114** includes a transmitter that transmits a signal to a transmit beamformer which in turn drives transducer elements within a transducer array to emit pulsed ultrasonic signals into a structure, such as a patient. The pulsed ultrasonic signals may be back-scattered from structures in the body, such as blood cells or muscular tissue, to produce echoes that return to the transducer elements. These echoes may then be converted into electrical signals by the transducer elements and the electrical signals are received by a receiver. The electrical signals representing the received echoes are sent to a receive beamformer that outputs ultrasound data. The processing device **2102** may be configured to process the ultrasound data from the ultrasound imaging device **2114** to generate ultrasound images for display on the display screen **2108**. The processing may be performed by, for example, the processor **2110**. The processor **2110** may also be adapted to control the acquisition of ultrasound data with the ultrasound imaging device **2114**. The ultrasound data may be processed in real-time during a scanning session as the echo signals are received. In some embodiments, the displayed ultrasound image may be updated a rate of at least 5 Hz, at least 10 Hz, at least 20 Hz, at a rate between 5 and 60 Hz, at a rate of more than 20 Hz. For example, ultrasound data may be acquired even as images are being generated based on previously acquired data and while a live ultrasound image is being displayed. As additional ultrasound data is acquired, additional frames or images generated from more-recently acquired ultrasound data are sequentially displayed. Additionally, or alternatively, the ultrasound data may be stored temporarily in a buffer during a scanning session and processed in less than real-time.

(112) Additionally (or alternatively), the processing device **2102** may be configured to perform any of the processes described herein (e.g., using the processor **2110**). For example, the processing device **2102** may be configured to automatically determine an anatomical feature being imaged and automatically select, based on the anatomical feature being imaged, an ultrasound imaging preset corresponding to the anatomical feature. As shown, the processing device **2102** may include one or more elements that may be used during the performance of such processes. For example, the processing device **2102** may include one or more processors **2110** (e.g., computer hardware processors) and one or more articles of manufacture that include non-transitory computer-readable storage media such as the memory **2112**. The processor **2110** may control writing data to and reading data from the memory **2112** in any suitable manner. To perform any of the functionality described herein, the processor **2110** may execute one or more processor-executable instructions stored in one or more non-transitory computer-readable storage media (e.g., the memory **2112**), which may serve as non-transitory computer-readable storage media storing processor-executable instructions for execution by the processor **2110**.

(113) In some embodiments, the processing device **2102** may include one or more input and/or output devices such as the audio output device **2104**, the imaging device **2106**, the display screen **2108**, and the vibration device **2109**. The audio output device **2104** may be a device that is configured to emit audible sound such as a speaker. The imaging device **2106** may be configured to detect light (e.g., visible light) to form an image such as a camera. The display screen **2108** may be configured to display images and/or videos such as a liquid crystal display (LCD), a plasma display, and/or an organic light emitting diode (OLED) display. The display screen **2108** may be a touch-sensitive display screen. The vibration device **2109** may be configured to vibrate one or more components of the processing device **2102** to provide tactile feedback. These input and/or output devices may be communicatively coupled to the processor **2110** and/or under the control of the processor **2110**. The processor **2110** may control these devices in accordance with a process being executed by the process **2110** (such as the processes shown in FIGS. 15-19). Similarly, the

processor **2110** may control the audio output device **2104** to issue audible instructions and/or control the vibration device **2109** to change an intensity of tactile feedback (e.g., vibration) to issue tactile instructions. Additionally (or alternatively), the processor **2110** may control the imaging device **2106** to capture non-acoustic images of the ultrasound imaging device **2114** being used on a subject to provide an operator of the ultrasound imaging device **2114** an augmented reality interface.

(114) It should be appreciated that the processing device **2102** may be implemented in any of a variety of ways. For example, the processing device **2102** may be implemented as a handheld device such as a mobile smartphone or a tablet. Thereby, an operator of the ultrasound imaging device **2114** may be able to operate the ultrasound imaging device **2114** with one hand and hold the processing device **2102** with another hand. In other examples, the processing device **2102** may be implemented as a portable device that is not a handheld device such as a laptop. In yet other examples, the processing device **2102** may be implemented as a stationary device such as a desktop computer.

(115) In some embodiments, the processing device **2102** may communicate with one or more external devices via the network **2116**. The processing device **2102** may be connected to the network **2116** over a wired connection (e.g., via an Ethernet cable) and/or a wireless connection (e.g., over a WiFi network). As shown in FIG. **21**, these external devices may include servers **2118**, workstations **2120**, and/or databases **2122**. The processing device **2102** may communicate with these devices to, for example, off-load computationally intensive tasks. For example, the processing device **2102** may send an ultrasound image over the network **2116** to the server **2118** for analysis (e.g., to identify an anatomical feature in the ultrasound) and receive the results of the analysis from the server **2118**. Additionally (or alternatively), the processing device **2102** may communicate with these devices to access information that is not available locally and/or update a central information repository. For example, the processing device **2102** may access the medical records of a subject being imaged with the ultrasound imaging device **2114** from a file stored in the database **2122**. In this example, the processing device **2102** may also provide one or more captured ultrasound images of the subject to the database **2122** to add to the medical record of the subject. For further description of ultrasound imaging devices and systems, see U.S. Patent Application Publication No. US20170360397A1 titled “UNIVERSAL ULTRASOUND DEVICE AND RELATED APPARATUS AND METHODS.”

(116) Various aspects of the present disclosure may be used alone, in combination, or in a variety of arrangements not specifically discussed in the embodiments described in the foregoing and is therefore not limited in its application to the details and arrangement of components set forth in the foregoing description or illustrated in the drawings. For example, aspects described in one embodiment may be combined in any manner with aspects described in other embodiments.

(117) The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

(118) The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified.

(119) As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to

which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified.

(120) Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

(121) The terms “approximately” and “about” may be used to mean within $\pm 20\%$ of a target value in some embodiments, within $\pm 10\%$ of a target value in some embodiments, within $\pm 5\%$ of a target value in some embodiments, and yet within $+2\%$ of a target value in some embodiments. The terms “approximately” and “about” may include the target value.

(122) Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having,” “containing,” “involving,” and variations thereof herein, is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

(123) Having described above several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be object of this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

Claims

1. A method for measuring a circumference of an anatomical structure, comprising: providing a handheld ultrasound probe for generating ultrasound images of the anatomical structure and providing a touchscreen display, being coupled to the handheld ultrasound probe for receiving ultrasound images from the handheld ultrasound probe for display, the touchscreen display having pixels for rendering a visible image of the ultrasound images from the handheld ultrasound probe, generating with the handheld ultrasound probe an ultrasound image that includes an image of the anatomical structure, displaying the ultrasound image that includes the image of the anatomical structure on the touchscreen display, providing a movable measurement tool having an ellipse with a major axis and a minor axis and a circumference and a visible representations of the major axis and a visible representation of the minor axis and a visible representation of a the circumference of the ellipse, displayed on the touchscreen display, and a first icon that maintains a fixed distance from the visible representations of the major axis and a second icon that maintains a fixed distance from the visible representations of the minor axis, allowing a user to drag the first icon across the touchscreen display and to drag the second icon across the touchscreen display to arrange the visible representations of the major axis to correspond to a length of the anatomical structure and the minor axis to correspond to a width of the anatomical structure displayed and to superimpose the circumference of the ellipse over the circumference of the anatomical structure while having the first icon spaced away by the fixed distance from the visible representation of the major axis of the ellipse and the second icon spaced away by the fixed distance from the visible representation of the minor axis of the ellipse, and determining a spatial length of the circumference of the ellipse by determining a number of the pixels used to display the circumference of the ellipse.
2. The method of claim 1, further including displaying on the touchscreen display text representative of a numerical value in centimeters of the spatial length of the circumference of the ellipse.
3. The method of claim 1, further including determining a spatial length of the major axis and a spatial length of the minor axis by determining the number of the pixels used to display the major and minor axes and determining a spatial area of the ellipse from the spatial lengths of the major and minor axes.

4. The method of claim 3, further including displaying on the touchscreen display text representative of a numerical value in squared centimeters of the area of the ellipse.
 5. The method of claim 3, wherein the determining of the spatial length of the major axis and the spatial length of the minor axis includes calculating fractional pixel results when a point on a respective one of the major axis and the minor axis falls between two adjacent pixels of the touchscreen display.
 6. The method of claim 1, further comprising providing at least one of the first and second icons with a rotation tool for rotating the visible representation of the circumference of the ellipse about an intersection point formed at an intersection of the major axis and the minor axis.
 7. The method of claim 6, wherein rotating the circumference of the ellipse includes allowing the user to rotate the major and minor axes of the ellipse to superimpose the circumference of the ellipse over a peripheral edge of the anatomical structure.
 8. The method of claim 1, further including providing a delete option for removing from the display the visual representation of the major and minor axes and the circumference of the ellipse.
 9. The method of claim 1, wherein the touchscreen display has the pixels arranged in an array of pixels where the array of pixels has an x-axis extending along a horizontal direction and a y-axis extending along a vertical direction and an origin point at an intersection of the x-axis and the y-axis, and each pixel in the array of pixels has a location determined by a distance measured from the x-axis and a distance measured from the y-axis.
 10. The method of claim 1, wherein allowing the user to drag the first icon to arrange the visible representations of the major axis includes allowing the user to touch a finger to the first icons and drag the finger a distance across the touchscreen display, to cause the length of the visible representation of the major axis to change by an amount proportionate to the distance which the finger dragged across the touchscreen display.
 11. The method of claim 10, wherein the length of the major axis changes by a multiple of the distance that the finger dragged the first icon across the touchscreen display.
 12. The method of claim 11, wherein the visible representation of the circumference of the ellipse is determined as a function of a Cartesian equation for the circumference of an ellipse.
 13. The method of claim 11, wherein the spatial length of the circumference of the ellipse is determined as a function of a Cartesian equation for the circumference of an ellipse and the spatial lengths of the major and minor axes.
 14. The method of claim 1, wherein displaying the ultrasound image on the touchscreen display, includes continuously updating the displayed ultrasound image.
 15. The method of claim 14, further including providing a freeze control to allow the user to control the display to hold the ultrasound image on the display by preventing the display from updating the ultrasound image.
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